

Math 104 — Homework 9

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Rudin Chapter 6

Problem 1 (6.16). For $1 < s < \infty$, define

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Prove that

$$(a) \quad \zeta(s) = s \int_1^{\infty} \frac{[x]}{x^{s+1}} dx,$$

$$(b) \quad \zeta(s) = \frac{s}{s-1} - s \int_1^{\infty} \frac{x-[x]}{x^{s+1}} dx,$$

where $[x]$ denotes the greatest integer $\leq x$. Prove that the integral in (b) converges for all $s > 0$.

Proof. (a) On each interval $[n-1, n)$ we have $[x] = n-1$, so

$$\int_1^{\infty} \frac{[x]}{x^{s+1}} dx = \sum_{n=2}^{\infty} \int_{n-1}^n \frac{n-1}{x^{s+1}} dx = \sum_{n=2}^{\infty} \frac{n-1}{s} \left(\frac{1}{(n-1)^s} - \frac{1}{n^s} \right).$$

Multiplying by s , we collect the coefficient of each $a_k := 1/k^s$ in the N th partial sum. The term $(n-1)(a_{n-1} - a_n)$ contributes $+(n-1)$ to a_{n-1} and $-(n-1)$ to a_n , so

$$\begin{aligned} S_N &= \sum_{n=2}^N (n-1)(a_{n-1} - a_n) \\ &= 1 \cdot a_1 + \sum_{k=2}^{N-1} [k - (k-1)] a_k - (N-1) a_N \\ &= \sum_{k=1}^{N-1} \frac{1}{k^s} - \frac{N-1}{N^s}. \end{aligned}$$

Since $s > 1$, $(N-1)/N^s \rightarrow 0$, so $S_N \rightarrow \zeta(s)$. Hence

$$\zeta(s) = s \int_1^{\infty} \frac{[x]}{x^{s+1}} dx.$$

(b) For $s > 1$, a direct computation gives

$$s \int_1^{\infty} \frac{x}{x^{s+1}} dx = s \int_1^{\infty} x^{-s} dx = \frac{s}{s-1}.$$

Splitting $x = [x] + (x - [x])$ and using part (a),

$$\frac{s}{s-1} = s \int_1^\infty \frac{[x]}{x^{s+1}} dx + s \int_1^\infty \frac{x - [x]}{x^{s+1}} dx = \zeta(s) + s \int_1^\infty \frac{x - [x]}{x^{s+1}} dx,$$

which rearranges to the desired identity.

Finally, for $s > 0$, since $0 \leq x - [x] < 1$,

$$0 \leq \int_1^\infty \frac{x - [x]}{x^{s+1}} dx \leq \int_1^\infty \frac{1}{x^{s+1}} dx = \frac{1}{s} < \infty,$$

so the integral in (b) converges for all $s > 0$. □

Problem 2 (6.17). Suppose α increases monotonically on $[a, b]$, g is continuous, and $g(x) = G'(x)$ for $a \leq x \leq b$. Prove that

$$\int_a^b \alpha(x)g(x) dx = G(b)\alpha(b) - G(a)\alpha(a) - \int_a^b G d\alpha.$$

Proof. Let $P = \{x_0, x_1, \dots, x_n\}$ be any partition of $[a, b]$. Since g is continuous and $G' = g$, the Mean Value Theorem applied to G on each $[x_{i-1}, x_i]$ gives a point $t_i \in (x_{i-1}, x_i)$ with

$$g(t_i) \Delta x_i = G(x_i) - G(x_{i-1}).$$

Form the Riemann sum for $\int_a^b \alpha g dx$ with this choice of tags:

$$S(P) := \sum_{i=1}^n \alpha(x_i) g(t_i) \Delta x_i = \sum_{i=1}^n \alpha(x_i) [G(x_i) - G(x_{i-1})].$$

Now we expand the telescoping sum $G(b)\alpha(b) - G(a)\alpha(a)$:

$$\begin{aligned} G(b)\alpha(b) - G(a)\alpha(a) &= \sum_{i=1}^n [\alpha(x_i)G(x_i) - \alpha(x_{i-1})G(x_{i-1})] \\ &= \sum_{i=1}^n \alpha(x_i) [G(x_i) - G(x_{i-1})] + \sum_{i=1}^n G(x_{i-1}) [\alpha(x_i) - \alpha(x_{i-1})] \\ &= S(P) + \sum_{i=1}^n G(x_{i-1}) \Delta \alpha_i. \end{aligned}$$

Rearranging,

$$S(P) = G(b)\alpha(b) - G(a)\alpha(a) - \sum_{i=1}^n G(x_{i-1}) \Delta \alpha_i.$$

We now let the mesh $\|P\| \rightarrow 0$.

Left side: Since α is monotone it is bounded and Riemann-integrable, and g is continuous, so $\alpha g \in R$ on $[a, b]$. The sum $S(P)$ is a Riemann sum for $\int_a^b \alpha g dx$ with tags t_i , so $S(P) \rightarrow \int_a^b \alpha(x)g(x) dx$.

Right side: G is an antiderivative of the continuous function g , hence G is continuous on $[a, b]$. Since α is monotone and G is continuous, $G \in R(\alpha)$, and the sum $\sum G(x_{i-1})\Delta\alpha_i$ is a Riemann–Stieltjes sum for $\int_a^b G d\alpha$ with left-endpoint tags, so it converges to $\int_a^b G d\alpha$.

Taking the limit in (*) gives

$$\int_a^b \alpha(x)g(x) dx = G(b)\alpha(b) - G(a)\alpha(a) - \int_a^b G d\alpha,$$

as desired. □

Problem 3 (6.18). Let $\gamma_1, \gamma_2, \gamma_3$ be curves in the complex plane, defined on $[0, 2\pi]$ by

$$\gamma_1(t) = e^{it}, \quad \gamma_2(t) = e^{2it}, \quad \gamma_3(t) = e^{2\pi it \sin(1/t)}.$$

Show that these three curves have the same range, that γ_1 and γ_2 are rectifiable, that the length of γ_1 is 2π , that the length of γ_2 is 4π , and that γ_3 is not rectifiable.

Proof. Same range. Each γ_j maps into $\{z \in \mathbb{C} : |z| = 1\}$ since $|e^{i\theta}| = 1$ for real θ . The image of γ_1 is the full unit circle (as t ranges over $[0, 2\pi]$, e^{it} traces out the circle once). For γ_2 , as t ranges over $[0, 2\pi]$, $2t$ ranges over $[0, 4\pi]$, traversing the circle twice; the range is still the full circle. For γ_3 , the exponent $2\pi t \sin(1/t)$ takes a continuum of real values, and by continuity plus the intermediate value theorem it sweeps through every angle modulo 2π , so the range is again the full unit circle.

Lengths of γ_1 and γ_2 . Both are C^1 , so their lengths are given by

$$\Lambda(\gamma_j) = \int_0^{2\pi} |\gamma_j'(t)| dt.$$

We compute $\gamma_1'(t) = ie^{it}$, so $|\gamma_1'(t)| = 1$ and $\Lambda(\gamma_1) = 2\pi$. Similarly $\gamma_2'(t) = 2ie^{2it}$, $|\gamma_2'(t)| = 2$, and $\Lambda(\gamma_2) = 4\pi$.

γ_3 is not rectifiable. Consider the argument $\theta(t) = 2\pi \cdot \frac{\sin(1/t)}{t}$, which oscillates unboundedly as $t \rightarrow 0^+$ since $|\sin(1/t)/t|$ becomes arbitrarily large. Choose a decreasing sequence $\{t_n\} \rightarrow 0$ in $(0, 2\pi]$ such that

$$\frac{\sin(1/t_n)}{t_n} = k_n \in \mathbb{Z},$$

with the k_n taking infinitely many distinct values; such a sequence exists by the intermediate value theorem applied on each interval where $\sin(1/t)/t$ sweeps through a large range of values.

Between consecutive t_n and t_{n+1} , the function $\sin(1/t)/t$ passes through every integer between k_n and k_{n+1} , so $\theta(t)$ sweeps through every multiple of 2π in that range. Between two consecutive such multiples, γ_3 traverses the full unit circle, contributing at least the diameter's worth (in fact the full circumference bound) to any polygonal approximation. Picking sample points at extremes of the oscillation, the polygonal length accumulates at least a fixed positive amount per “sweep,” and since there are infinitely many sweeps as $t \rightarrow 0$, the total polygonal length is unbounded. Hence

$$\Lambda(\gamma_3) = \sup_P \sum_i |\gamma_3(t_i) - \gamma_3(t_{i-1})| = +\infty,$$

so γ_3 is not rectifiable. □

Rudin Chapter 7

Problem 4 (7.1). *Prove that every uniformly convergent sequence of bounded functions is uniformly bounded.*

Proof. Let $\{f_n\}$ be a sequence of bounded functions on a set E converging uniformly to f . For each n , since f_n is bounded, let

$$M_n := \sup_{x \in E} f_n(x), \quad M'_n := \inf_{x \in E} f_n(x),$$

so $M'_n \leq f_n(x) \leq M_n$ for all $x \in E$.

By uniform convergence, pick N such that

$$|f_n(x) - f(x)| < 1 \quad \text{for all } n \geq N \text{ and all } x \in E.$$

In particular, $|f_N(x) - f(x)| < 1$, so

$$|f(x)| \leq |f_N(x)| + 1 \leq \max(|M_N|, |M'_N|) + 1$$

for all $x \in E$; thus f is bounded. For every $n \geq N$,

$$|f_n(x)| \leq |f(x)| + 1 \leq \max(|M_N|, |M'_N|) + 2.$$

There are only finitely many indices $n < N$, and each such f_n is bounded by $\max(|M_n|, |M'_n|)$. Setting

$$M := \max\left(\max(|M_N|, |M'_N|) + 2, \max_{1 \leq n < N} \max(|M_n|, |M'_n|)\right),$$

we obtain $|f_n(x)| \leq M$ for all n and all $x \in E$. Hence $\{f_n\}$ is uniformly bounded. $\square$

Problem 5 (7.3). *Construct sequences $\{f_n\}, \{g_n\}$ which converge uniformly on some set E , but such that $\{f_n g_n\}$ does not converge uniformly on E (of course, $\{f_n g_n\}$ must converge on E).*

Proof. Take $E = \mathbb{R}$ and define

$$f_n(x) = \frac{1}{n}, \quad g_n(x) = x.$$

$f_n \rightarrow 0$ **uniformly on $\mathbb{R}$** : Indeed

$$\sup_{x \in \mathbb{R}} |f_n(x) - 0| = \frac{1}{n} \rightarrow 0.$$

$g_n \rightarrow g$ **uniformly on $\mathbb{R}$** : Here $g(x) = x$ and $g_n = g$ for every n , so

$$\sup_{x \in \mathbb{R}} |g_n(x) - g(x)| = 0$$

for all n . Trivially uniform.

$f_n g_n$ does not converge uniformly: The product is $f_n(x)g_n(x) = x/n$, which converges pointwise to 0 for every fixed $x \in \mathbb{R}$. But

$$\sup_{x \in \mathbb{R}} \left| \frac{x}{n} - 0 \right| = \sup_{x \in \mathbb{R}} \frac{|x|}{n} = +\infty$$

for every n , so the sup-norm error does not tend to 0. Hence $\{f_n g_n\}$ does not converge uniformly on $\mathbb{R}$. $\square$

Problem 6 (7.4). Consider

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{1+n^2 x}.$$

For what values of x does the series converge absolutely? On what intervals does it converge uniformly? On what intervals does it fail to converge uniformly? Is f continuous wherever the series converges? Is f bounded?

Proof. **Domain / absolute convergence.** The n th term $u_n(x) = \frac{1}{1+n^2 x}$ is undefined when $1+n^2 x = 0$, i.e. at $x = -1/n^2$. At $x = 0$, each term equals 1 and $\sum 1$ diverges. For every other x , as $n \rightarrow \infty$,

$$\left| \frac{1}{1+n^2 x} \right| \sim \frac{1}{n^2 |x|},$$

so by comparison with $\sum 1/n^2$ the series converges absolutely. Hence the series converges absolutely on

$$D := \mathbb{R} \setminus \left(\{0\} \cup \{-1/n^2 : n \geq 1\} \right).$$

Uniform convergence via the M -test. Fix any closed interval $I \subset D$ bounded away from 0 and from every $-1/n^2$. Then there exists a constant $C > 0$ such that $|1+n^2 x| \geq Cn^2$ for all $x \in I$ and all sufficiently large n (because $|x|$ is bounded below by some $\delta > 0$ on I , giving $|1+n^2 x| \geq n^2 \delta - 1 \geq Cn^2$ for large n). Thus

$$\sup_{x \in I} \left| \frac{1}{1+n^2 x} \right| \leq \frac{1}{Cn^2},$$

and $\sum 1/(Cn^2) < \infty$. By the Weierstrass M -test, the series converges uniformly on I .

Failure of uniform convergence on $(0, \infty)$. For any N , consider the tail

$$R_N(x) = \sum_{n=N+1}^{\infty} \frac{1}{1+n^2 x}.$$

As $x \rightarrow 0^+$, each term tends to 1, so in particular $R_N(x) \geq \frac{1}{1+(N+1)^2 x} \rightarrow 1$ as $x \rightarrow 0^+$. Hence $\sup_{x>0} R_N(x) \geq 1$ for every N , so the convergence is not uniform on $(0, \infty)$. The same reasoning shows non-uniform convergence on any interval whose closure contains 0 or any $-1/n^2$.

Continuity. Each term $u_n(x)$ is continuous on D . The series converges uniformly on every closed interval $I \subset D$ bounded away from 0 and the $-1/n^2$, so by Theorem 7.12 the

sum f is continuous on every such I . Since continuity is local and every point of D has such a neighborhood, f is continuous on all of D .

Boundedness. f is *not* bounded. As $x \rightarrow 0^+$, every term $\frac{1}{1+n^2x} \rightarrow 1$, and for any N ,

$$f(x) \geq \sum_{n=1}^N \frac{1}{1+n^2x} \rightarrow N \quad \text{as } x \rightarrow 0^+.$$

Since N is arbitrary, $f(x) \rightarrow +\infty$ as $x \rightarrow 0^+$, so f is unbounded on D . $\square$

Problem 7 (7.5). Let

$$f_n(x) = \begin{cases} 0 & (x < \frac{1}{n+1}), \\ \sin^2 \frac{\pi}{x} & (\frac{1}{n+1} \leq x \leq \frac{1}{n}), \\ 0 & (\frac{1}{n} < x). \end{cases}$$

Show that $\{f_n\}$ converges to a continuous function, but not uniformly. Use the series $\sum f_n$ to show that absolute convergence, even for all x , does not imply uniform convergence.

Proof. **Pointwise limit is 0.** Fix $x \in \mathbb{R}$.

- If $x \leq 0$, then $x < 1/(n+1)$ for every $n \geq 1$, so $f_n(x) = 0$ for all n .
- If $x > 1$, then $x > 1/n$ for every $n \geq 1$, so $f_n(x) = 0$ for all n .
- If $0 < x \leq 1$, pick N with $1/N \leq x$. Then for every $n \geq N$ we have $x \geq 1/N \geq 1/n$, so $x > 1/n$ fails only on $[1/(n+1), 1/n]$; once n is large enough that $1/n < x$, we have $f_n(x) = 0$.

In every case, $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$. The support of f_n is $[1/(n+1), 1/n]$, an interval whose position shrinks toward 0 as $n \rightarrow \infty$, so for each fixed x eventually x lies outside the support and the error $|f_n(x) - 0| = 0$. Hence $f_n \rightarrow 0$ pointwise, and the limit $f \equiv 0$ is continuous.

Not uniform. Inside the support $[1/(n+1), 1/n]$, the function $\sin^2(\pi/x)$ attains the value 1: choose $x_n \in [1/(n+1), 1/n]$ with $\pi/x_n = \pi/2 + k\pi$ for an appropriate integer k , giving $\sin^2(\pi/x_n) = 1$. Such x_n exists for every n because the interval $[1/(n+1), 1/n]$ under the map $x \mapsto \pi/x$ becomes $[n\pi, (n+1)\pi]$, which has length π and hence contains a point of the form $\pi/2 + k\pi$. Therefore

$$\sup_{x \in \mathbb{R}} |f_n(x) - 0| = 1 \quad \text{for every } n,$$

which does not tend to 0. Hence $f_n \rightarrow 0$ is *not* uniform.

The series $\sum f_n$. For any fixed x , at most one of the disjoint supports $[1/(n+1), 1/n]$ contains x , so at most one term $f_n(x)$ is nonzero. In particular

$$\sum_{n=1}^{\infty} |f_n(x)| \leq 1 < \infty,$$

so the series converges absolutely for every $x \in \mathbb{R}$.

However, the convergence is not uniform. Let $S(x) = \sum_{n=1}^{\infty} f_n(x)$ and let $S_N(x) = \sum_{n=1}^N f_n(x)$ be the N th partial sum. Then

$$S(x) - S_N(x) = \sum_{n=N+1}^{\infty} f_n(x),$$

and by the same argument as above, this tail agrees with $\sin^2(\pi/x)$ on each interval $[1/(n+1), 1/n]$ for $n \geq N+1$, hence attains the value 1 on those intervals. Therefore

$$\sup_{x \in \mathbb{R}} |S(x) - S_N(x)| = 1 \quad \text{for every } N,$$

which never tends to 0. So $\sum f_n$ converges absolutely but not uniformly, showing that absolute convergence does not imply uniform convergence. $\square$

AI Use Disclaimer: Claude (Anthropic) was used in the preparation of this assignment. Claude served solely as a transcription and formatting tool, taking verbal dictation of my solutions and converting them into L^AT_EX. Claude did not provide answers, solve problems, or generate proofs. It was used only as a guide to help structure my own reasoning, never as a solver.